

ESIM Bench Test

Evaporative System Integrity Module: vacuum switch, seal and relief check

2018 Alfa Romeo Stelvio 2.0T | VIN ZASFAKPN5J7B88115 | about 142,290 km

Why this test

The ECU has stored **P0455** (large leak), **P0456** (small leak) and **P0440** (EVAP system) repeatedly. Every inexpensive cause has already been ruled out. The ESIM is the part that tells the ECU whether the EVAP system is sealed. If its switch never closes, or its internal seal leaks, the ECU records a leak even when every hose is tight, and all three codes can come from that one fault. This test decides whether the ESIM is at fault before any part is bought.

Already checked on this car

Check	Result
EVAP air filter	Clear and clean
Recirculation-line mid-point quick-connect	Seated and latched
Liquid fuel at the canister	None
Refuelling habit	Stops at first click; no topping off
Filling symptoms, last 2 fills	No hiss, no smell, no slow filling
Water ingress at the canister	None found
Purge valve (engine bay)	Replaced with a new unit; clicks
Fuel cap	New
Check-engine light	Off. The ECU stored and confirmed the codes without requesting it (status 0x4D, light bit clear), so the light is not a guide for this fault.

Source note. FCA bulletin 9100469 confirms the ESIM is a separate part from the vapour canister and says to replace only the ESIM when the fault is in the ESIM. No factory bench-test procedure, switch threshold or connector pinout for this car was obtainable. The steps below are general bench practice for this type of vacuum switch. **Compare against a new ESIM** wherever a number matters.

Safety

- The ESIM and canister hold fuel vapour. Work outdoors or in a ventilated bay, away from sparks, flames and running engines.
- Ignition OFF before unplugging the connector.
- Apply vacuum and pressure **gently**. The switch acts at a very small vacuum; a hard pump stroke can damage the diaphragm.
- Never use shop air or any regulated air line on the ESIM.

Tools

- Multimeter set to ohms or continuity.
- Hand vacuum pump with a short length of rubber hose.

- A low-range vacuum gauge or a simple water manometer (a clear U-tube of water). A hand pump's inHg dial is too coarse to see where this switch closes.
- Rubber vacuum caps, a flat trim tool, and a torch.
- Optional but best: a new ESIM to test side by side.

Procedure

Step 1. Access and inspect the connector

- Remove the driver-side rear wheel liner. The ESIM sits on the vent side of the canister.
- With the ignition off, unplug the ESIM connector.
- Look for green corrosion, water, pushed-back or bent pins, and a loose lock. The ESIM sits in road spray; a poor connection alone makes the ECU read the switch as open.

Step 2. Remove the ESIM or free its ports

- Remove it, or disconnect both hoses. Before disconnecting, **mark which port faces the canister** and which faces the fresh-air filter.

Step 3. Switch at rest

- Put the meter across the switch terminals with the ESIM at atmospheric pressure.
- **Expected: open** (OL or no continuity).
- If the connector has more than two pins, find the pair whose reading changes in Step 4.

Step 4. Vacuum test on the canister-side port

Fresh-air side left open to air. Pump vacuum into the canister-side port slowly.

- **Switch closes:** the meter snaps to near 0 ohms at a small vacuum. This is the ECU's 'system sealed' signal.
- **Hold:** stop pumping. The vacuum should hold steady and the switch stay closed. This tests the internal seal.
- **Release:** let the vacuum go. The switch should open again.
- Note the gauge reading at which it closes, and whether the vacuum falls while holding.

Step 5. Pressure relief check

- Apply gentle low pressure to the canister-side port.
- **Expected:** it vents through to the fresh-air side and does not hold. This confirms the ESIM is not stuck shut, which matches the normal fill-ups on this car.

Step 6. Tap and wiggle

- With the meter connected and the switch held closed by vacuum, tap the body lightly and wiggle the terminals. Watch for the reading flickering or dropping out.

Reading the result

What you see	What it means	Next step
Switch never closes	Failed switch. The ECU sees a leak on a tight system, which fits all three codes.	Replace the ESIM only (TSB 9100469).
Closes, but the vacuum bleeds away	Internal seal leaking: a real leak inside the ESIM.	Replace the ESIM only.
Closes only at a much higher vacuum than a new unit	Sluggish switch; can fail the engine-off test intermittently.	Replace the ESIM.

What you see	What it means	Next step
Reading flickers when tapped	Failing contacts.	Replace the ESIM; check the harness connector too.
Behaves like a new unit on every step	ESIM exonerated. The leak is elsewhere.	Run the smoke test by section: canister first.

Record sheet

Fill this in during the test. These values go into the cuore evidence gate: the readings are the measurement, and a clean pass on every step is the disconfirming test that clears the ESIM.

Item	Result
Date / odometer	
Connector condition	
Step 3: resting reading (should be open)	
Step 4: vacuum at which the switch closed	
Step 4: vacuum held? (drop over 1 minute)	
Step 4: switch reopened on release?	
Step 5: vents under light pressure?	
Step 6: any flicker when tapped?	
New-unit comparison (if available)	
Conclusion	

After the test

- **If the ESIM failed:** replace only the ESIM, clear the codes, then drive several times with fuel between about 15 and 85 percent, including cold starts. Confirm with cuore's verify_repair, which reads whether each leak test has re-run and passed. The check-engine light cannot confirm it on this car.
- **If the ESIM passed:** it is exonerated. Run the smoke test by section (docs/research/EVAP_SMOKE_TEST.md), introducing smoke at the canister purge port rather than the fuel filler, so the new cap is tested where it sits.
- A road test alone cannot confirm a small-leak repair (TSB 18-048-23).